

BIOS 794: Support Vector Machines

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October 6, 2025

Outline

Introduction

No perfect classifier

Support Vector Machines

Overview

- ▶ Motivation for margin-based classifiers
- ▶ Support Vector Classifier (SVC)
- ▶ Support Vector Machine (SVM) with kernels
- ▶ Hyperparameters and tuning
- ▶ When and why SVMs work well

Notation and problem set up.

- ▶ Let $\mathbf{X} \in \mathbb{R}^p$ and $Y \in \{-1, 1\}$, and we want to predict what class each observation comes from $\mathbf{X}$.
- ▶ **Separating hyperplane/boundary:** exists if a line (or more generally a plane of some dimensions) can perfectly separate the data into classes.

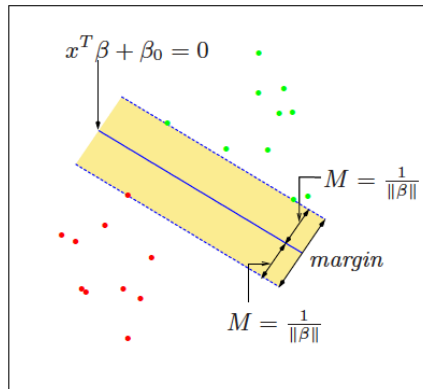
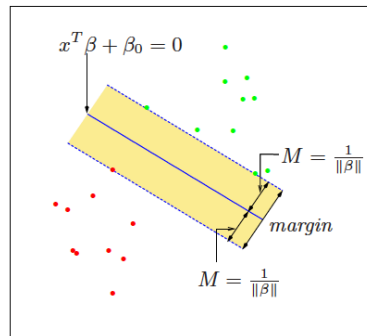


Figure: A linear separating hyperplane/boundary.

Motivation: Classification with Margins

- ▶ Many classifiers (e.g., logistic regression) find a separating hyperplane/boundary.
- ▶ When there is a separating hyperplane, there will actually be infinitely many.
- ▶ Can we find the one with the *largest margin*?
- ▶ Margin: distance from the boundary to the nearest observation.



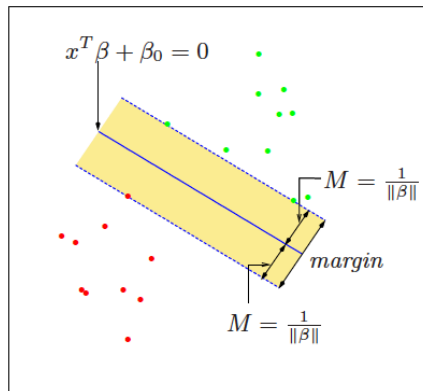


Figure: **Support** points are those that lie on the dotted line.

Support Vector Classifier (SVC)

- ▶ If the data are linearly separable, then there exists a β_0 and β such that

$$\beta_0 + \mathbf{X}'_i \beta \geq +1 \quad \text{if } y_i = +1$$

$$\beta_0 + \mathbf{X}'_i \beta \leq -1 \quad \text{if } y_i = -1$$

- ▶ The points that lie on these lines are referred to as **support points**.
- ▶ This leads to the **Hard-margin SVC** optimization problem.

$$\min_{\beta, \beta_0} \frac{1}{2} \|\beta\|^2 \quad \text{subject to } y_i(\beta^T \mathbf{x}_i + \beta_0) \geq 1$$

Note: only the support points impact the estimates of β_0 and β .

Support Vector Classifier (SVC)

- ▶ What if there are no separating hyperplanes?
- ▶ Be slack!! (i.e., use a “soft margin” solution)
- ▶ **Soft-margin SVC:** Allow slack variables ξ_i :

$$\min_{\beta, \beta_0, \xi} \frac{1}{2} \|\beta\|^2 + C \sum \xi_i \quad \text{subject to } y_i(\beta^T x_i + \beta_0) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

This allows points to be on the inside of the margin (or even misspecified).

- ▶ C : margin vs misclassification tradeoff.
- ▶ Support vectors: observations on the margin or misclassified.

Support Vector Classifier (SVC)

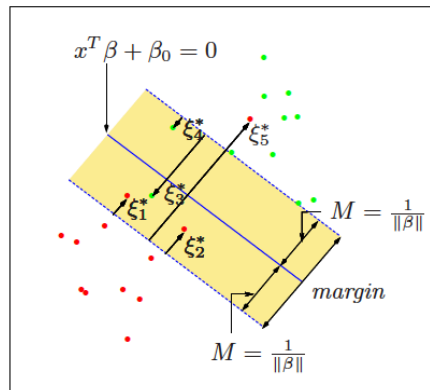
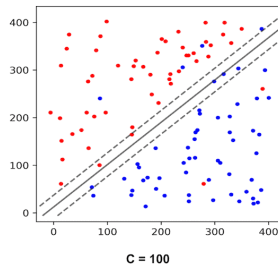
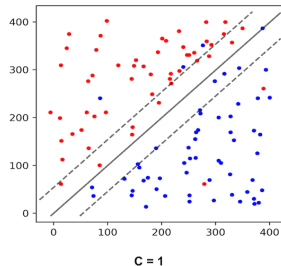


Figure: Here, there does not exist a hyperplane that perfectly separates the data.

Hyperparameter: Cost (C)

- ▶ **Low C:** Large margin, more misclassifications.
- ▶ **High C:** Tight margin, fewer misclassifications.

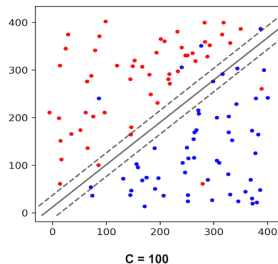
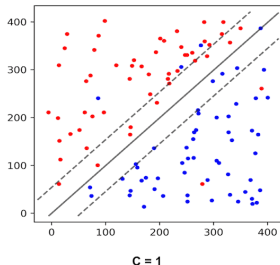
SVM Parameter C



Hyperparameter: Cost (C)

- ▶ **Low C**: Large margin, more misclassifications.
- ▶ **High C**: Tight margin, fewer misclassifications.
- ▶ Tune with cross-validation.

SVM Parameter C



Support Vector Machines: Introduction

- ▶ Non-linear support vector machines are what most people consider to be support vector machines.
- ▶ Linear support vector machines are usually referred to as support vector classifiers (terminology we'll use here), but that is not universal.
- ▶ The linear support vector classifier solves

$$\min_{\beta_0, \beta} \left\{ \frac{1}{2} \|\beta\|^2 + C \sum_{i=1}^n \xi_i \right\}$$

such that $\xi_i \geq 0$ and $y_i(\beta_0 + \mathbf{x}'_i \beta) \geq 1 - \xi_i$ for all i .

Support Vector Machines

- ▶ We could, of course, consider transformations of $\mathbf{x}$ such as the basis functions we discussed previously.
- ▶ For example, suppose we use $h(\mathbf{x}) = \{h_1(\mathbf{x}), h_2(\mathbf{x}), \dots, h_M(\mathbf{x})\}$ as the variables for our support vector classifier, and

$$\hat{f}(\mathbf{x}) = h(\mathbf{x})' \hat{\beta} + \hat{\beta}_0$$

is our classifier.

- ▶ Sometimes by projecting the data into higher dimensions, we can make non-separable data separable.

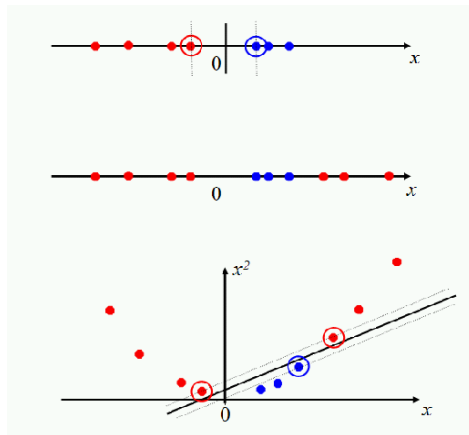


Figure: Here $\mathbf{x} = x$ and $h(\mathbf{x}) = \{x, x^2\}$.

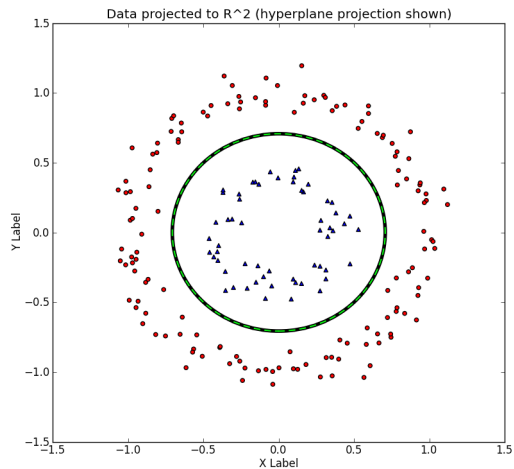
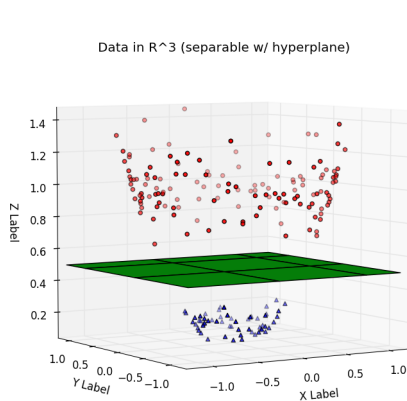


Figure: Here $\mathbf{x} = (x_1, x_2)$ and $h(\mathbf{x}) = \{x_1^2, \sqrt{2}x_1x_2, x_2^2\}$.

Systematic functional mappings

- ▶ Suppose $h \in \mathcal{H}$ is a general mapping of a p -dimensional covariate vector $\mathbf{x}$.
- ▶ For example, a quadratic with all two-way interactions

$$h(\mathbf{x}) = (x_1^2, \dots, x_p^2, x_1x_2, x_1x_3, \dots, x_1x_p, \dots, x_{p-1}x_p)$$

- ▶ The dimension of h blows up!
- ▶ Calculating $\sum_i \sum_j h(\mathbf{x}_i)^\top h(\mathbf{x}_j)$ gets very expensive.
- ▶ Also, what if you wanted to use an infinite-dimensional $h(\mathbf{x})$?

Kernal Trick

- ▶ **Kernal Trick**, replace

$$h(\mathbf{x}_i)^\top h(\mathbf{x}_j) = K(\mathbf{x}_i, \mathbf{x}_j)$$

where $K(\cdot, \cdot)$ is a Kernal function.

- ▶ The choice of $K(\cdot, \cdot)$ implicitly determines h and $\mathcal{H}$.
- ▶ Given a Kernel, we don't have to pick h .
- ▶ Calculating $K(\mathbf{x}_i, \mathbf{x}_j)$ is much easier then calculating $h(\mathbf{x}_i)^\top h(\mathbf{x}_j)$

Examples of Kernels

- **Polynomial of degree d :** $K(\mathbf{x}_i, \mathbf{x}_j) = (1 + \langle \mathbf{x}_i, \mathbf{x}_j \rangle)^d$
- Suppose $\mathbf{x} = (x_1, x_2)$ with $d = 2$, we have

$$\begin{aligned} K(\mathbf{x}_i, \mathbf{x}_j) &= (1 + \mathbf{x}_i^\top \mathbf{x}_j)^d \\ &= (1 + x_{i1}x_{j1} + x_{i2}x_{j2})^2 \\ &= 1 + 2x_{i1}x_{j1} + 2x_{i2}x_{j2} + (x_{i1}x_{j1})^2 + (x_{i2}x_{j2})^2 + 2x_{i1}x_{j1}x_{i2}x_{j2} \\ &= h(\mathbf{x}_i)^\top h(\mathbf{x}_j) \end{aligned}$$

when $h(\mathbf{x}) = (1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, x_2^2, \sqrt{2}x_1x_2)$

Examples of Kernels

- ▶ **Linear:** $K(\mathbf{x}_i, \mathbf{x}_j) = \mathbf{x}_i^\top \mathbf{x}_j$
- ▶ **Polynomial of degree d :** $K(\mathbf{x}_i, \mathbf{x}_j) = (1 + \mathbf{x}_i^\top \mathbf{x}_j)^d$
- ▶ **Radial basis:** $K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2)$.
- ▶ **Neural network:** $K(\mathbf{x}_i, \mathbf{x}_j) = \tanh(\kappa_1 \mathbf{x}_i^\top \mathbf{x}_j + \kappa_2)$.
- ▶ Each of these kernels has a corresponding (sometimes infinite-dimensional) $h(\mathbf{x})$, where perfect separation is often achievable.
- ▶ C plays a big role.
 - ▶ A large C will discourage any positive ξ_i , and overfit the model with a wiggly boundary.
 - ▶ A small C will encourage a small value of $\|\beta\|$ resulting in smooth $f(x)$

Hyperparameter: Kernel and Gamma

- ▶ **Kernel choice:** depends on data geometry.
- ▶ **Gamma (γ)** (RBF/poly): controls kernel flexibility.
 - ▶ Small γ : smoother decision boundary.
 - ▶ Large γ : tight, complex boundary.
- ▶ Tune γ and C jointly.

Tuning SVMs

- ▶ Use `caret`, `e1071`, or `mlr3` in R.
- ▶ Scale features before training.
- ▶ Grid search or random search with CV.
- ▶ Evaluate using AUC, F1, accuracy.

Pros of SVM

- ▶ **Effectiveness in high-dimensional spaces:** SVMs are particularly effective in cases where the number of dimensions is greater than the number of samples.
- ▶ **Versatility:** Custom kernels can be used for the decision function.
- ▶ **Accuracy:** SVMs generally provide accurate classifiers.

Cons of SVM

- ▶ **Complexity and memory:** SVMs can be painfully inefficient to train. They use a significant amount of memory.
- ▶ **No direct probability estimation:** SVMs do not directly provide probability estimates; these are calculated using an expensive five-fold cross-validation.
- ▶ **Difficult (Black-box) Interpretation:** The results of predictions can be very difficult to interpret compared to some other algorithms.
- ▶ **Tuning Required:** The success of an SVM is heavily dependent on the correct tuning of the hyperparameters. Hyperparameter tuning is critical.
- ▶ **Poor performance with noise:** If the data is noisy or overlaps, it can lead to a decrease in performance as SVMs are sensitive to the noise in the dataset.

Summary

- ▶ SVC maximizes margin; soft-margin allows flexibility.
- ▶ SVMs with kernels handle nonlinear problems.
- ▶ Tune C and kernel parameters carefully.
- ▶ Visualization helps develop intuition.